Status	Finished
Started	Tuesday, 20 October 2026, 12:58 PM
Completed	Tuesday, 20 October 2026, 1:02 PM
Duration	3 mins 39 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It requires the list to be sorted ✓
c. It does not work with negative numbers
d. It is slower than linear search for small lists

Question **2**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search does not require dividing the list
d. Binary search can find multiple instances of the target element

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list is unsorted
c. When the list is sorted ✓
d. When the list contains duplicate elements

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The search is considered unsuccessful ✓
- b. The first element is returned
- c. An error is raised
- d. The last element is returned

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Quick search
- b. Linear search ✓
- c. Bubble search
- d. Merge search

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. The list must be sorted
- b. There should be direct access to the middle element in any sublist
- c. There must be a mechanism to delete and/or insert elements in the list ✓
- d. Number values should only be present

Number values should only be present

Question **7**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the right sublist ✓
- b. In the middle of the list
- c. At the beginning of the list
- d. In the left sublist

Question **8**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O(n log n)
c. O👎
d. O(1) ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓
b. O(log n)
c. O(1)
d. O(n log n)

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By comparing each element sequentially
b. By using a hash function
c. By dividing the list length by two ✓
d. By starting from the first element

Question **11**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n-1
c. log n
d. n ✓

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search can be applied to both sorted and unsorted lists. ✓
c. Linear search requires the list to be sorted.
d. Linear search divides the list into halves.

Question **13**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 89 and 99
c. 90 and 99
d. 90 and 94 ✗

Question **14**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search
b. Both (1) & (3)
c. Hash search
d. Linear search ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The search stops
c. The list is sorted
d. The search continues in the left or right sublist ✓